Eclipse-Timing Analysis of TIC 233720276 Using TESS Observations

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Abstract. Based on the observations by the Transiting Exoplanet Survey Satellite (TESS), this paper investigates the eclipse timing of the binary star system TIC 233720276. A total of 41 sectors of 120 second light curves were analysed. An eclipse-shape model was fitted to each primary and secondary eclipse to determine the time of minimum light intensity, and a total of 407 primary eclipses and 415 secondary eclipses were selected for analysis. The first set of eclipses had a refined period of 2.198388719804815 days and an O-C standard deviation of 1.810 seconds. The measured first-eclipse times were in agreement with the constant-period prediction. However, after removing the average offset at phase 0.5, the secondary eclipse was no longer about 88 seconds before the primary eclipse but now occurred about 78 seconds after. The rate of increase was approximately 31.4 seconds, per year. The results show that the relative position or measured shape of the secondary eclipse changed during the TESS observations. Apsidal motion is a possible reason; however, star spots, changes in eclipse shape and other remaining data effects have not yet been excluded.

1 Introduction

Eclipsing binaries are multiple-star systems that orbit each other, and the plane of rotation is nearly parallel to the line of sight from the observer. Periodically, the components of an eclipsing binary block each other and thus show regular changes in light intensity. The above multi-star systems can be used to infer the orbital period of a star and thus provide support for studying stellar evolution and internal processes [1]. Alterations in the orbital period of eclipsing binaries are also due to various physical reasons. The reasons for orbital changes are mainly as follows: mass transfer from one star to another, mass loss due to stellar winds, tidal interaction between the two stars, the light-time effect caused by a third body, and magnetic activity cycles [2]. Accurate measurement of the period change can thus provide support for binary evolution, mass-transfer history and the possible presence of a third body.

To investigate the alteration in the orbital period of an eclipsing binary over time, an O-C diagram is commonly used. Given an initial observed minimum (zero epoch) T0 and period P, the predicted eclipse time for cycle E is Tcalc=T0+P*E. The difference O-C=Tobs-Tcalc

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is plotted against the reference time to find a possible period. A horizontal trend is a stable period; a parabolic trend is a changing period; and a sinusoidal trend may be caused by a light-travel-time effect from a third body or cyclic magnetic activity [2]. Ways to interpret O-C diagrams differ. The traditional method of piecewise linear or quadratic fitting is relatively restrictive; a newer continuous method can be used to model the entire O-C curve and is more physically reasonable. This paper studies the traditional mode of O-C diagram interpretation.

Space-based Photometric Missions have improved eclipse-timing accuracy. TESS by NASA is another high-precision survey covering nearly the entire sky [3]. Recently, some research has combined TESS data with historical records to conduct high-precision studies of eclipsing binaries. Yang and Wang studied 1,495 new TESS minima from four binaries and detected light-travel-time signals in all four systems [4]. Zhang and Zhang found cyclic O-C oscillations in the two short-period systems that may be caused by third bodies [5]. Wolf and others combined TESS eclipse timings with historical observations to improve apsidal-motion measurements for three eccentric eclipsing binaries [6]. Marcadon and Prša have developed an automated TESS full-frame-image timing method to find new triple-system candidates on a large scale [7]. Borkovits et al. later found 168 hierarchical triple candidates, including 66 high-confidence ones, in the northern TESS continuous viewing zone [8].

This study selects the eclipsing binary TIC 233720276 and uses TESS light curves from MAST to construct O-C diagrams for its primary and secondary eclipses. Test whether a constant-period ephemeris can fit the observations and determine if the two eclipse types have the same timing characteristics.

2 Method and Theory

2.1 Target selection and the TESS data

The O-C method will be used to find the time of the eclipse here. Calculate the average of all eclipses and compare it to the time predicted by a constant-period model. From the catalog of 4,584 TESS eclipsing binaries published by Prša et al. [1], a target was selected. Filter the catalog for stars with periods between 0.4 and 3.0 days, TESS magnitudes brighter than 12, morphology values between 0.15 and 0.50, and observations in at least three sectors. TIC 233720276 was selected because it is bright, has many TESS observations, and has a deep eclipse.

Data retrieval, analysis, and visualization were performed in Python using Lightkurve, Astropy, Astroquery, SciPy, NumPy, and Matplotlib. Lightkurve was used to query the Mikulski Archive for Space Telescopes for TIC 233720276 [9]. Only the official SPOC light curves with a measurement interval of 120 seconds were used for the search. The original FITS files were employed. TESS stores its times as BTJ1D, which is BJD_TDB minus 2, 457, 000, so 2, 457, 000 was added to each time. No second correction was required because the TESS pipeline had already shifted the observations to the centre of the Solar System.

A measurement was retained only if its time and sector were valid, its quality flag was zero, and its brightness and positive brightness error were available. Each sector was normalized by dividing its brightness and error by the sector's median brightness. Set the Normal Brightness close to 1.0. Both SAP and PDCSAP light curves were verified. PDCSAP is frequently used in many areas to remove the common instrumental effect. SAP was used in Sectors 18, 23, 56, 60, 73, 77 and 78 because the PDCSAP corrections visibly altered the eclipse shape in these sectors.

No missing measurement records occurred. Otherwise, the distance between adjacent points was too large; otherwise, it would be flagged. The Gaps were re-assessed after each

eclipse was observed. Thus, the fabricated points inside the gap would not create a false minimum or shift the eclipse time.

2.2 Eclipse timing determination and orbital ephemeris refinement

A Box Least Squares search was employed to find a starting period [10]. To reduce the search space, only every fifth valid point was selected. The first search used 20,000 periods between 0.10 and 2.00 days and five possible eclipse lengths. A second search tested another 20,000 periods around the best result. The largest value was doubled when the folded curve indicated that BLS had treated the primary and secondary eclipses as repeated drops. The time of the deeper eclipse was taken as the first T_0.

All the usable measurements were folded at the beginning to place the observations from different orbits on the same phase scale. Phase zero was the anticipated time of the first eclipse. The Stage is as follows:

$$\phi = \left[\left(\frac{t - T_0}{P} + 0.5\right) mod\ 1\right] - 0.5 \tag{1}$$

About the first and last eclipses were split into 600 equal-phase bins. Separate measurements near the secondary eclipse were taken and divided into 500 bins. The median brightness was chosen for each bin because it is less sensitive to outliers than the mean.

The median points were fitted to a simplified eclipse model as shown in Mikulášek [11]. Change the centre, normal brightness, nearby slope, depth, width and roundness of the model until it matches the binned eclipse. SciPy was used to test the values using a least-squares fit, and a soft-L1 method was used to reduce the impact of outliers [12]. The model was only used to find the time and not to calculate the actual size of the stars. Templates and start times were used to predict all possible primary and secondary eclipses. The primary predictions returned integer cycle numbers, and the secondary predictions used half-integer cycles because they were first expected halfway between two primary eclipses. In each fit, the general eclipse shape was kept constant, and only the centre time, normal brightness, brightness slope and depth were allowed to vary. Maintaining a single form kept the times relatively close.

Coverage tests were used to avoid false time errors due to incomplete eclipses. Each candidate needs to have been measured before and after prediction, have sufficient phase coverage on both sides, and no large gap in the fitting window. Points more than five standard deviations away from the first fit were excluded prior to the final weighted fit. A fit was rejected if its time or depth exceeded the boundary, or if the timing error was invalid. Reasons for all the refusals were saved. The accepted primary eclipse times were used to perform a weighted straight-line fit. Shorter error times were given more weight. Calculation based on cycle 481, which is close to the middle of the observation, and the formula was used:

$$T_{calc}(E) = T_{ref} + P_{ref}\left(E - E_{ref}\right) \tag{2}$$

A five-sigma outlier check was repeated up to five times. The resulting period and reference time were then used to calculate an improved prediction:

$$T_{calc}(E) = 2459770.341881524 + 2.198388719804815(E - 481)\mathrm{BJD_{TDB}}$$

Calculate the O-C value of each valid eclipse by subtracting the predicted time from the observed time:

$$(O - C)_E = T_{obs}(E) - T_{calc}(E) \tag{3}$$

The secondary eclipse occurred before exact phase 0.5 by a small, almost constant amount. The median offset was excluded from both the primary and secondary changes before comparison. Therefore, the comparison showed a shift in the position of the secondary eclipse rather than just an average early location. The median primary and secondary O-C values for each sector have also been computed.

3 Results

3.1 TESS observations and fully folded curve

The target filters had 117 unmatched stars. TIC 233720276 has a TESS magnitude of 8.49298 and a catalog period of 2.1983907126 days. Its Right Ascension is 276.35773854 degrees, its Declination is +62.94038818 degrees, and its Morphology value is 0.473452. The archive search found 41 matching light-curve products in Sectors 15-26, 40, 47-60, and 73-86. The above observations are from 2019-2024. After quality inspection, 659 and 643 measurements were left: 97,606 from SAP and 562,037 from PDCSAP. The code also found 249 gaps. No whole sector was deleted just because a gap appeared somewhere in it.

The BLS search first found 1.099194700972 days. This was about half the catalog period because the two eclipses were considered the same repeated drop. Doubling the value resulted in a starting period of 2.198389401944 days. The starting primary time was T0 = 2458712.9141491633 BJD_TDB. Folding the data at full-period separation resulted in a deeper primary eclipse at phase 0 and a secondary eclipse near phase 0.5, as shown in Figure 1.

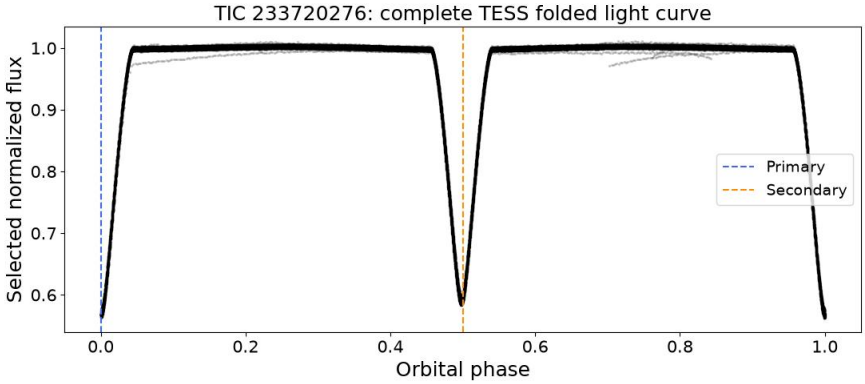


Fig. 1. Complete folded TESS light curve for TIC 233720276. The dashed lines are the first and second eclipses.

3.2 Primary and secondary eclipse results

The first template had 196,178 measurements in the range of -0.15 to 0.15. The fitted centre moved by 209.712 seconds, and the template time was 2458712.9165763874 BJD_TDB. The model had a depth of 0.433286 and a root-mean-square error of 0.002549.

The first search returned 447 items from cycles 0 to 887. A total of 407 candidates passed all the tests, and 40 were disqualified. Seventeen of the rejections were due to large gaps, and most of the others lacked sufficient measurements or coverage on one side. The median timing error was 2.400 seconds, and the range of errors was 2.045-5.578 seconds.

The first set of primary O-C values showed a downward trend; thus, the starting period was slightly too long. The weighted fit retained all 407 approved times. The refined period was 2.198388719804815 days, about 0.05894 seconds shorter per orbit than the starting value. The fitted reference time was 2459770.341881524 BJD_TDB at cycle 481. The internal uncertainty is 0.1205 seconds and the period uncertainty is 0.000414 seconds.

A total of about 5.33 years, from August 2019 to December 2024. The refined O-C values had a standard deviation of 1.810 seconds and ranged from -11.464 to 6.677 seconds. They were close to zero and no longer formed a distinct slope. Figure 2 compares the preliminary and refined primary O-C diagrams.

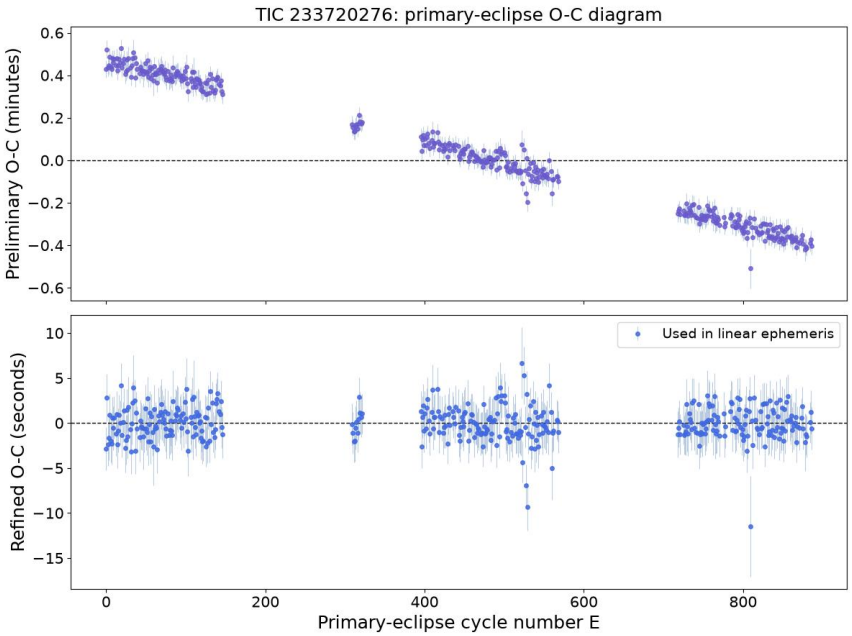


Fig. 2. Primary-eclipse O-C Diagram. The upper panel is the starting period, and the lower panel shows the remaining time differences after refining the period.

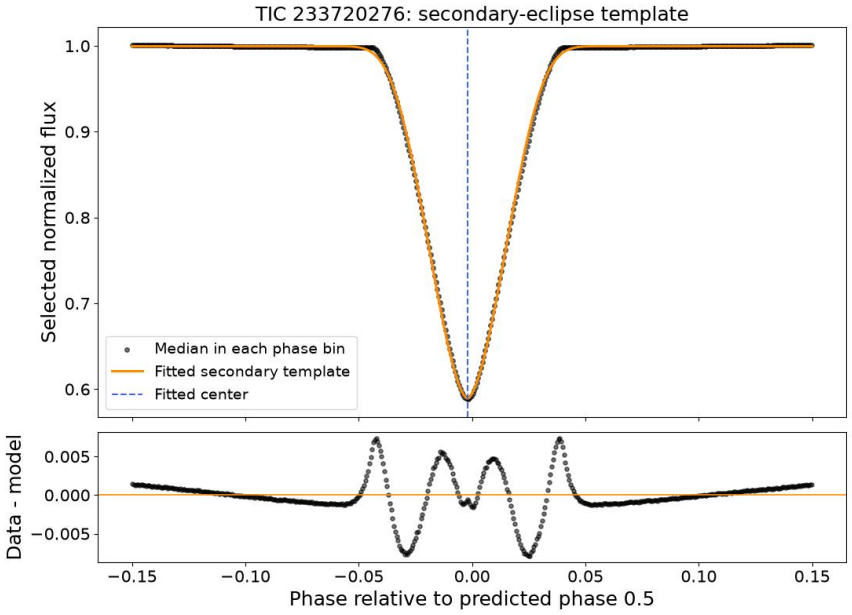


Fig. 3. Secondary-eclipse template. The black dots are the median brightness values, the orange line is the fitted model, and the lower panel shows the residuals after fitting.

The second template had 197,779 measurements and 500 phase bins. The centre was at phase -0.002006 relative to phase 0.5. This is the starting offset of -381.08 seconds. The fitted depth was 0.410605, the width was 0.018673 in phase, and the root-mean-square difference was 0.002601. The fitted template and the remaining residuals are shown in Figure 3.

There were 451 secondary fitting attempts. Among them, 415 passed the tests and 36 failed. The main reasons for rejection were missing coverage on one side and large gaps. The median secondary timing error was 3.621 seconds, and the acceptable range of errors was 3.006-5.622 seconds. The secondary eclipse did not occur at phase 0.5 precisely. The median measured offset was -379.292 seconds, so it appeared about 6.32 minutes before the midpoint prediction. After subtracting the constant offset, the individual secondary O-C values were in the range of -94.945 to +79.224 seconds and had a standard deviation of 55.304 seconds.

The medians of the primary sector ranged from -1.18 to 1.53 seconds and were nearly uncorrelated. The median of the centered secondary sector was -89.51 seconds in Sector 15 and +77.65 seconds in Sector 86. The difference between the secondary and primary was now about 166.6 seconds after rising from -88.38 seconds. A straight line fitted to the 41 sector differences increased by about 31.4 seconds per year and had $R^2 = 0.999$. Figure 4 is the main and secondary timing patterns.

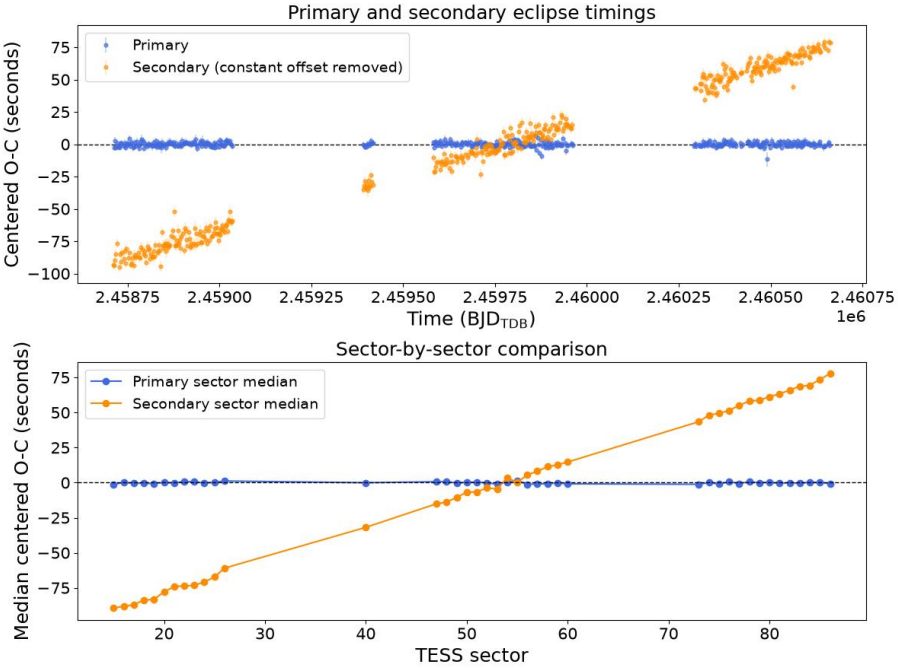


Fig. 4. Comparison of the Primary and Secondary Eclipse Times. The constant secondary offset at phase 0.5 was not included in the comparison. The lower panel shows the median values of these areas.

4 Discussion

The main eclipse times are well-described by a single constant period over the 5.33 years of TESS observations. The standard deviations of these are 1.810 seconds, and the medians of the sectors do not show a significant increase or decrease. Therefore, the main data do not offer strong support for a rapid change in the orbital period. A much slower change may still occur because the current time period only covers a few years.

The second eclipses are not the same. If only the size of the entire orbit were changing, then both types of eclipses would move together in relation to the same prediction. Instead, the main eclipse was still close to zero, and the position of the secondary eclipse changed gradually with respect to it. The change of about 166.6 seconds is also relatively large

compared with the median secondary timing error of 3.621 seconds. Random fitting error is therefore unlikely to account for the entire change.

One possible reason is apsidal motion. If the orbit is slightly elliptical and not perfectly circular, then the direction of the ellipse will also vary slowly. It will change the time difference of the primary and secondary eclipses, but the mean orbital period will be almost unchanged. The secondary eclipse occurs about 6.32 minutes before the exact phase of 0.5, and thus the orbit or eclipse shape is not perfectly symmetrical. Constant velocity refers to motion in a circle at a steady speed. However, the O-C diagram alone cannot show apsidal motion.

A Change in Eclipse Shape is another possible reason. Starspots are relatively dark and will thus appear dimmer in the light of an eclipse. A symmetric template may be too far to the left or right in the middle. The residual panels of both templates have small repeating patterns, and the accepted secondary fits have a median reduced chi-squared value of about 23.7. Therefore, the simple model cannot show all the small details and does not have a good fit for the eclipse shape.

They are relatively easy to compare because the main and secondary measurements were taken with the same telescope, sectors and general data processing. Many instrumental problems will affect both sets in the same way. SAP and PDCSAP data were still combined by sector quality, and there were long gaps between groups of sectors. Use only one kind of flux and different eclipse models for the analysis to see if the trend is still the same. The current work only covers TESS observations. The historical times mentioned in the introduction have not been added yet, so this paper cannot show how the system has performed over the past few decades. Historical primary and secondary eclipse times, future TESS observations, and a physical model of the stars can be used to test apsidal motion, starspots, a third body, and other explanations. The safest current result is that the secondary eclipse has shifted away from the stable primary timing during the TESS observations.

5 Conclusion

Measure the eclipse times of TIC 233720276 in 41 TESS sectors in this study. The 407 primary eclipses produced a refined period of 2.198388719804815 days and were close to a linear prediction, with an O-C standard deviation of 1.810 seconds. Therefore, a single constant period can account for most of the eclipses over the 5.33-year TESS baseline. On the other hand, after subtracting the mean early offset of the 415 accepted secondary eclipses, their time of occurrence shifted from about 88 seconds before the primary pattern to about 78 seconds after, or about 31.4 seconds per year. As the primary and secondary eclipses did not occur simultaneously, it is unlikely that only a change in the orbital period has taken place. Apsidal motion is a possible reason; that is, a rotating eccentric orbit can alter the distance between the two eclipses without significantly changing the period. However, this is not the end. Starspots, changing eclipse shapes, a symmetric template, mixed SAP and PDCSAP measurements, and gaps in the observing sector can also shift the measured secondary centre.

Future work will add historical minima, repeat the analysis with different eclipse models, and separate SAP-only and PDCSAP-only data sets to determine whether the trend continues in the later TESS sectors. Physical light-curve and radial-velocity modelling can determine the necessary orbital eccentricity for apsidal motion analysis. Continuously observe to determine if it is a stable orbit or a temporary surface phenomenon. The strongest current conclusion is that the secondary eclipse has shifted systematically with respect to the stable primary timing, and its physical cause has not yet been identified.

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